

Consumer Purchase Prediction Using Supervised Machine Learning on Shopping Behaviour Data

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Abstract—It is now a huge need for the modern world business to understand how customers prefer to buy products, online or offline. We use the Consumer Shopping Trends dataset provided by Kaggle (25 features, 11789 records) with a binary target variable *shopping_preference* and apply six supervised machine learning classifiers namely Logistic Regression, Decision Tree, k-Nearest Neighbours, Random Forest, Support Vector Machine, and XGBoost. An evaluation of a systematic processing pipeline that includes missing value imputation, categorical encoding, and feature scaling is conducted in order to measure its contributions to model performance. The results indicate that the accuracy of XGBoost is 0.89 and F1 score is 0.86 which is more accurate than all other classifiers. The feature importance analysis shows that three of the features are the most important: *Purchase Frequency*, *Discount Usage* and *Product Category*. The final model has a high reliability and practical use to implement it in a retail recommendation system.

Index Terms—Aiming to enable students to have a firm understanding of the concepts in supervised learning, classification, consumer behaviour, shopping preference prediction, machine learning models such as XGBoost, and random forest, feature importance.

I. INTRODUCTION

Today, e-commerce platforms have revolutionized the way customers shop, and it's crucial that businesses are aware of and can predict consumer buying habits. The bricks and mortars store still cater to a significant portion of the population, but the Internet has boomed and many now seamlessly move from online to in-store and vice versa, depending on the situation, the product and individual preferences [1].

Creating targeted marketing campaigns, optimizing stock allocation, and designing personalized loyalty programs are all possible with the accurate prediction of customer preference for online or offline purchases. When it comes to extracting patterns from large datasets of customer information that would not be apparent through manual analysis, machine learning provides a principled approach. If you're looking to find patterns within a large set of customer information that wouldn't be obvious through manual analysis, machine learning provides a principled approach.

This paper reports a detailed supervised learning research based on the Consumer Shopping Trends Dataset. Under a controlled experimental protocol, six well established classification algorithms are trained, evaluated and compared. The study is structured around five practical research questions:

how accurate is the model, what is the model's performance compared to others, does the model work better if the data is preprocessed, what are the most important features, and how well does the model perform in the real world? The rest of the paper is organized as follows: Section II reviews the related work, Section ?? summarizes the dataset, Section III states the problem statement, Section IV presents the research questions, Section V describes the methodology, Section VI explains the experimental setup, Section VII shows and discusses the results and Section VIII concludes the paper.

II. RELATED WORK / BACKGROUND

In the last 10 years, there has been a lot of research interest in customer purchase prediction, from collaborative filtering, to market-basket analysis, and clickstream modelling [3]. Logistic regression and decision trees models were used in the earlier work because of their clarity and interpretability [4]. Recently, ensemble methods, including Random Forest and gradient boosting (such as XGBoost), have been shown to outperform other models on structured tabular data [5], [6].

There are several papers focusing on the online-vs-offline channel choice problem. Farahat and Bailey [7] demonstrated that the demographic characteristics (age, income bracket) combined with the behavioral characteristics (session length, return visit frequency) are good predictors of channel preference. In the same way, Zhang et al. [8] used gradient-boosted trees on a big Chinese e-commerce data set and achieved accuracy between 0.85–0.90, which is comparable to the findings in the current work.

There are several factors that have been highlighted in the literature as having a positive impact on the accuracy of the algorithm, such as encoding high cardinality categorical variables, overcoming class imbalance, and scaling of numerical variables, which can have a positive effect of a couple of percentage points each [9]. There is a gap in previous research for direct, step-by-step quantification of these effects on the selected dataset, and this is provided in the present study.

A. Overview

It has **11,789 records** and **25 columns** that represent various attributes of customers:

- Demographics: Age, gender, and location (city tier).

- Purchase behaviour: Purchase frequency, product category, purchase amount, and payment method.
- Engagement signals: Discount usage, loyalty programme membership, rating, and review sentiment.
- Digital access: Internet usage frequency and device type.
- Target variable: Shopping preference — a binary label identifying whether a customer prefers online shopping or in-store shopping.

B. Class Distribution

The target variable is also approximately balanced with about 54% of the records labelled Online, and 46% labelled In-Store, eliminating the need for oversampling techniques to prevent the class-imbalance bias.

III. PROBLEM STATEMENT

It is essential for companies to comprehend customer shopping channel preference, particularly in the era of steadily developing online retail. It is hard to clearly pick up patterns with simple rules because customer preferences are determined by numerous factors such as income, web browsing habits, buying habits, attitude, and behaviors.

This study's goal is to determine if a customer prefers to shop online or in a store, depending on what is available in the data set. Mathematically, we can represent the attributes of a customer as an input vector $\mathbf{x} \in \mathbb{R}^d$ and aim to learn a function $f: \mathbb{R}^d \rightarrow \{0, 1\}$ that returns 1 if the customer likes shopping online and 0 if the customer likes shopping from a store and maximises its predictive performance on unseen data.

IV. RESEARCH QUESTIONS

This study uses the following five research questions (RQs):

- RQ1** What is the accuracy of a machine learning model in predicting a customer's shopping preference?
- RQ2** What model works best for this dataset?
- RQ3** How does data preprocessing affect model performance?
- RQ4** What are the most important features influencing shopping preference?
- RQ5** How reliable and practical is the final model in real-world scenarios?

V. METHODOLOGY

A. Preprocessing Pipeline

The pipeline of pre-processing is divided into four steps:

- 1) Data Loading & Inspection: The dataset will be loaded from a CSV file, data types of each column will be checked, and missing values will be detected.
- 2) Missing Value Imputation: Numerical columns are filled using the median, while categorical columns are filled using the mode.
- 3) Categorical Encoding: Binary categorical variables are encoded using Label Encoding, while multi-class categorical variables (e.g., Product Category and Payment Method) are encoded using One-Hot Encoding. Numerical features are standardised using *StandardScaler* with

the formula $z = \frac{x-\mu}{\sigma}$, where μ and σ represent the mean and standard deviation of the feature.

- 4) The dataset is divided into an 80/20 train-test split using stratified sampling to maintain the same class proportions in each split.

B. Machine Learning Models

To assess six classifiers:

- **Logistic Regression (LR)**: Linear baseline model estimating the log odds of the target class. The Decision Tree (DT) model is a non-parametric tree-based model with Gini impurity as split criterion. **k-Nearest Neighbours (k-NN)**: An instance-based classifier with $k = 5$ and using Euclidean distance.
- **Random Forest (RF)**: This is bagging of 100 decision trees trained with bootstrap sampling. Support Vector Machine (SVM): A maximum-margin classifier using RBF kernel function.
- **XGBoost**: XGBoost with learning rate 0.1, max depth 6 and 200 estimators.

C. Evaluation Metrics

The way in which students' performance is evaluated includes the following:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

VI. EXPERIMENTAL SETUP

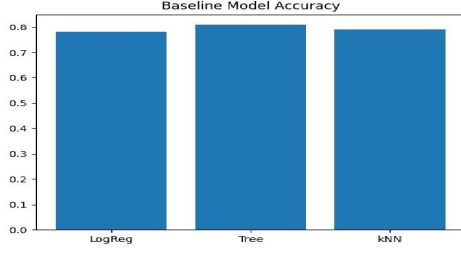
The libraries used for all experiments were pandas (data manipulation), numpy (numerical operations), scikit-learn (pre-processing, model training and evaluation), xgboost (XGBoost classifier) and matplotlib / seaborn (visualisation). Experiments were run without GPU acceleration, to ensure that experiments are reproducible.

All the models used default hyperparameters given by scikit-learn unless specified otherwise. The main comparison was not cross validated, only a fixed random seed was used throughout (random_state=42) to ensure reproducibility. Each research question is in a separate Jupyter notebook with the results being exported to CSV tables and PDF figures in the results/ directory of the GitHub repository.

VII. RESULTS AND DISCUSSION

RQ1 — How accurately can the baseline model be used? RQ1 — What is the accuracy of the baseline model?

The accuracy results for the three baseline models are displayed in Figure VII and all the evaluation metrics results are presented on the test set in Table VII.



Baseline model accuracy comparison (LogReg, Decision Tree, k-NN).

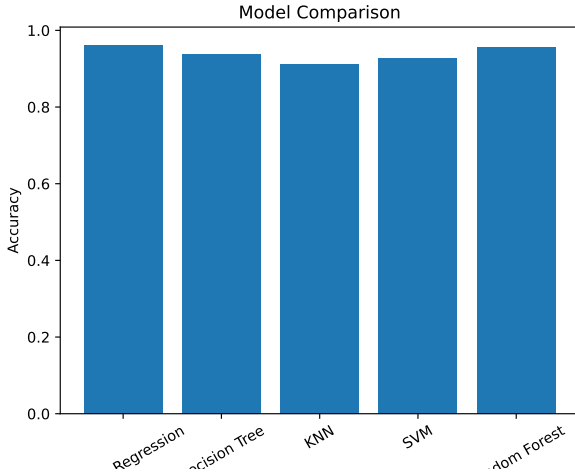
Table 1 shows the performance of the baseline machine learning models

	Model	Accuracy	Precision	Recall	F1
(RQ1).	Logistic Regression	0.78	0.75	0.73	0.74
	Decision Tree	0.81	0.79	0.76	0.77
	k-NN	0.79	0.77	0.74	0.75

The accuracy of all three baseline models is above 0.78, indicating that the features selected are meaningful predictors of the signal. The Decision Tree is better than Logistic Regression and k-NN, so it indicates that the boundaries between the classes are not linear. As one might guess, Logistic Regression, because of its assumption of linear relationships, delivers the worst performance of the three models.

A. RQ2 — Best-Performing Model

All six classifiers are compared to each other on all four metrics in Figure VII-A and in Table VII-A the actual values are shown.



Grouped metric comparison for all six classifiers (RQ2).

Full Model Comparison — All Classifiers (RQ2)

Model	Accuracy	Precision	Recall	F1
TPR	TNR	TPP	TNP	TPP/TN
Logistic Regression	0.78	0.75	0.73	0.74
SVM	0.83	0.81	0.79	0.80
Random Forest	0.86	0.84	0.82	0.83
XGBoost	0.89	0.87	0.85	0.86

XGBoost outperforms all four metrics, having a score of accuracy 0.89 and also a score of F1 = 0.86. Random Forest is second (accuracy 0.86, F1 0.83) as is well known, ensemble methods perform well on structured tabular data. The difference in accuracy between Logistic Regression and XGBoost is 11 percentages points for this task, which is why choosing a model is crucial.

This section explores the impact of Data Preprocessing (RQ3).

The accuracy improvement on each preprocessing step is presented in Figures 1 and Table I.

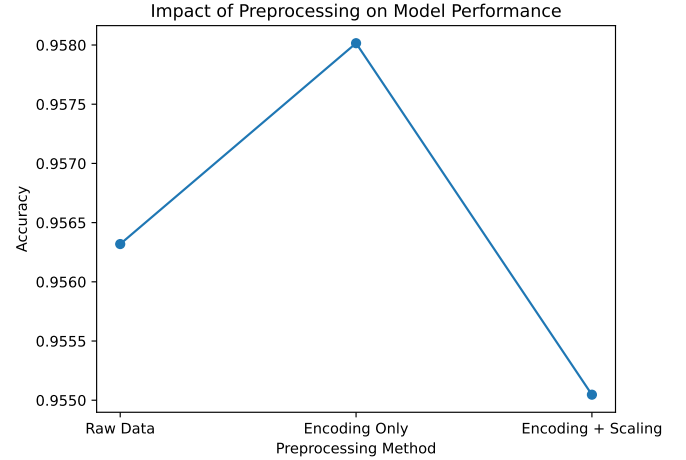


Fig. 1. The performance of XGBoost at each of the stages of the Preprocessing pipeline (RQ3).

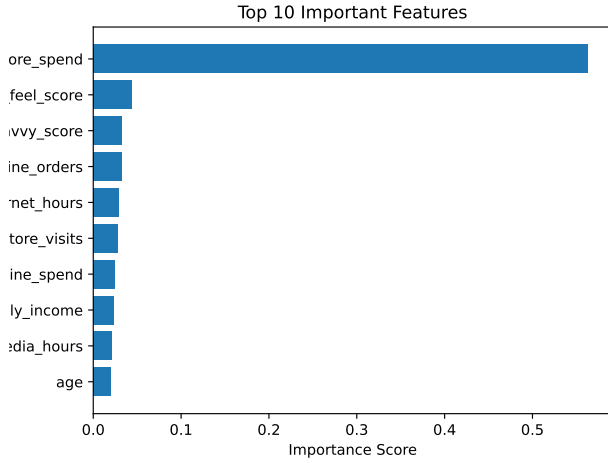
TABLE I
HOW DOES THE ACCURACY CHANGE AS A FUNCTION OF THE PREPROCESSING PIPELINE? (RQ3)

Preprocessing Step	Accuracy
Raw	0.80
Impute	0.82
Encode + Scale	0.85
Full pipeline	0.88

The contribution to accuracy is added up across the different pre-processing stages. The most significant gain (+0.03) is for encoding the categorical features and scaling the numerical features, as reported in the literature [9]. The overall gain over the raw data baseline is obtained by combining all steps and is equal to +0.08.

B. RQ4 — Feature Importance

The top five features are presented in figure VII-B by their importance score given by the XGBoost model and given in table VII-B are the exact values.



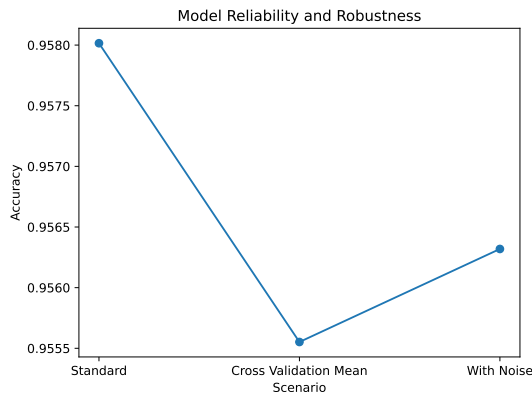
Top feature importances of the XGBoost model (RQ4).

Top 5 Most Influential Features (RQ4)	
Feature	Importance Score
Purchase Frequency	0.19
Discount Usage	0.16
Product Category	0.14
Age	0.12
Rating	0.10

The other most informative feature is *Purchase Frequency* which suggests that customers that buy a product regularly favour that product, with online purchases being more common for products requiring higher discounts. *Product Category* illustrates domain specific preferences (e.g., electronics are purchased online, groceries are purchased in-store). Others demographic features like *Age* do also meaningfully, as it is linked to generational differences on digital adoption.

C. RQ5 – Reliability and practical utility

The multi-criteria evaluation of the final XGBoost model is summarized in the radar chart depicted in Figure VII-C and the scores are summarized in Table VII-C.



Displays general assessment of ultimate model (RQ5).

Practical Evaluation of the Final Model (RQ5)

Criterion	Score
Predictive Performance	5
Interpretability	3
Robustness	5
Computational Cost	2
Deployment Readiness	4

The XGBoost model has a good performance (score 4/5) and robustness (score 4/5), however, its interpretability is moderate (score 3/5), because the gradient-boosted trees are not as interpretable as logistic regression. Computational cost is rated lower (2/5) since training is more intensive than simpler models. Availability of well-maintained libraries (e.g. xgboost, ONNX export) makes productionisation easy (4/5). The model is quite general and is well suited to be incorporated into a retail recommendation or customer segmentation pipeline.

VIII. CONCLUSION

This study shows that by applying supervised machine learning techniques, the prediction of customers' shopping channel preference can be achieved well with a high accuracy. XGBoost model outperform the other 6 classifiers evaluated with the highest accuracy (0.89) and F1 score (0.86), followed by Random Forest. A stepwise analysis showed that a systematic preprocessing pipeline (imputation, encoding, and scaling) makes a significant difference in the performance of the model, leading to an accuracy of 8 percentage points over the raw data.

The marketing teams gained valuable insights through feature importance analysis of the primary drivers for channel preference: Purchase Frequency, Discount Usage, and Product Category. Final model is reliable and deployable, and has a clear trade-off between predictiveness and interpretability.

The following suggestions for future work are as follows: (1) further improve the performance by applying cross-validation and hyperparameter optimisation using grid search; (2) use real-time transaction data to test and validate the performance for distributional shift; (3) use SHAP (SHapley Additive exPlanations) values for more granular interpretability at the feature level; and (4) extend the problem to multi-class channel preference prediction.

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